

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0135

Hell Creek 138 kV
March 26, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0135. The Interconnection Customer proposes a 211.1 MW natural gas combined cycle facility interconnecting at the Hell Creek 138 kV substation in the New Jersey area. The request was received on June 07, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 91 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,509,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Crows Landing to Fairview 138 kV
Contingency: Windhub to Hell Crk 138 kV
Loading: 150.4% (Rating: 212 MVA)
DFAX: 21.8%
Pre-project loading: 128.7%
2. Facility: Tomac to JacksonSquare 138 kV Line 1
Contingency: Centralf to Windhub 138 kV Line
Loading: 144.8% (Rating: 289 MVA)
DFAX: 31.4%
Pre-project loading: 121.8%
3. Facility: Mount Laurel to Lumberton 138 kV
Contingency: Hell Crk to Windhub 138 kV
Loading: 139.3% (Rating: 239 MVA)
DFAX: 21.4%
Pre-project loading: 120.4%
4. Facility: Tomac to Jackson Square Substation 138 kV Circuit 2
Contingency: Windhub to Centralf 138 kV line
Loading: 139.0% (Rating: 301 MVA)
DFAX: 31.4%
Pre-project loading: 117.0%
5. Facility: Little Rock Arch St to La Crosse 138 kV Line
Contingency: Corinth to Lolo 138 kV
Loading: 134.8% (Rating: 222 MVA)
DFAX: 7.5%
Pre-project loading: 127.7%

6. Facility: Little Rock Arch St to Camden 138 kV
Contingency: Corinth to Lolo 138 kV
Loading: 133.4% (Rating: 190 MVA)
DFAX: 7.5%
Pre-project loading: 125.1%
7. Facility: Fairview to Dunbar 138 kV
Contingency: Windhub to Hell Crk 138 kV line
Loading: 130.3% (Rating: 226 MVA)
DFAX: 21.8%
Pre-project loading: 110.0%
8. Facility: Mount Laurel to Border 138 kV line
Contingency: Centralf to University Wps 138 kV
Loading: 129.7% (Rating: 233 MVA)
DFAX: 19.5%
Pre-project loading: 112.0%
9. Facility: Shepherd Center Substation to Lolo 138 kV Line
Contingency: Oakland X to La Crosse 138 kV L
Loading: 128.5% (Rating: 220 MVA)
DFAX: 7.4%
Pre-project loading: 121.5%
10. Facility: Shepherd Center Substation to Burholme 138 kV
Contingency: La Crosse to Oakland X 138 kV
Loading: 127.1% (Rating: 202 MVA)
DFAX: 7.4%
Pre-project loading: 119.4%
11. Facility: Dunbar to Dpc Gradiot 138 kV Line
Contingency: Hell Crk to Windhub 138 kV line
Loading: 114.5% (Rating: 237 MVA)
DFAX: 21.8%
Pre-project loading: 95.2%
12. Facility: Border to Marlton 138 kV
Contingency: Hell Creek to Framingham 138 kV
Loading: 101.4% (Rating: 477 MVA)
DFAX: 33.0%
Pre-project loading: 86.8%
13. Facility: Jackson Square to Reno 138 kV line
Contingency: Hell Creek to Windhub 138 kV
Loading: 101.3% (Rating: 829 MVA)
DFAX: 56.8%
Pre-project loading: 86.8%
14. Facility: Jackson Square to Country Club 2 Tap 138 kV
Contingency: Centralf Substation to Windub 138 kV line
Loading: 101.2% (Rating: 571 MVA)
DFAX: 38.9%
Pre-project loading: 86.8%
15. Facility: Mount Laurel to University Wps 138 kV
Contingency: Greenwood to Centralf 138 kV
Loading: 101.1% (Rating: 459 MVA)
DFAX: 29.4%
Pre-project loading: 87.6%
16. Facility: Country Club 2 Tap to Dillion 138 kV
Contingency: Centralf Substation to Windhub 138 kV

Loading: 100.9% (Rating: 583 MVA)

DFAX: 38.9%

Pre-project loading: 86.8%

17. Facility: Greenwood to Marlton 138 kV

Contingency: Framingham to Hell Creek 138 kV

Loading: 100.7% (Rating: 506 MVA)

DFAX: 33.0%

Pre-project loading: 86.9%

18. Facility: University Wps to Centralf 138 kV

Contingency: Centralf to Greenwood 138 kV Line

Loading: 100.4% (Rating: 484 MVA)

DFAX: 29.4%

Pre-project loading: 87.6%

19. Facility: Lumberton to Vicent 138 kV

Contingency: Hell Crk to Windhub 138 kV

Loading: 100.4% (Rating: 361 MVA)

DFAX: 21.4%

Pre-project loading: 87.9%

20. Facility: Windhub Substation to Dillion 138 kV

Contingency: Centralf to Windhub 138 kV

Loading: 100.2% (Rating: 616 MVA)

DFAX: 38.9%

Pre-project loading: 86.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Crows Landing to Fairview 138 kV	Rebuild line with higher-capacity conductor	\$1,072,000	12% (\$132,000)
Tomac to JacksonSquare 138 kV Line 1	Rebuild line with higher-capacity conductor	\$967,000	12% (\$119,000)
Mount Laurel to Lumberton 138 kV	Rebuild line with higher-capacity conductor	\$2,516,000	12% (\$310,000)
Tomac to Jackson Square Substation 138 kV Circuit 2	Rebuild line with higher-capacity conductor	\$783,000	12% (\$97,000)
Little Rock Arch St to La Crosse 138 kV Line	Rebuild line with higher-capacity conductor	\$3,683,000	12% (\$456,000)
Little Rock Arch St to Camden 138 kV Line	Rebuild line with higher-capacity conductor	\$1,500,000	12% (\$186,000)
Dunbar to Fairview 138 kV	Rebuild line with higher-capacity conductor	\$2,199,000	12% (\$271,000)
Mount Laurel to Border 138 kV line	Rebuild line with higher-capacity conductor	\$4,887,000	12% (\$603,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lolo to Shepherd Center 138 kV	Rebuild line with higher-capacity conductor	\$6,131,000	12% (\$759,000)
Shepherd Ctr to Burholme 138 kV	Rebuild line with higher-capacity conductor	\$1,430,000	12% (\$177,000)
Dunbar to Dpc Gradiot 138 kV	Rebuild line with higher-capacity conductor	\$6,558,000	12% (\$809,000)
Border to Marlton 138 kV	Reconductor line	\$407,000	12% (\$50,000)
Jackson Square to Reno 138 kV line	Reconductor line	\$2,555,000	12% (\$315,000)
Jackson Square to Country Club 2 Tap 138 kV	Reconductor line	\$570,000	12% (\$70,000)
Mount Laurel Substation to University Wps 138 kV	Reconductor line	\$2,920,000	12% (\$360,000)
Country Club 2 Tap Substation to Dillion 138 kV line	Reconductor line	\$853,000	12% (\$105,000)
Greenwood to Marlton 138 kV	Reconductor line	\$1,185,000	12% (\$146,000)
University Wps to Centraif 138 kV	Reconductor line	\$253,000	12% (\$31,000)
Lumberton to Vincent 138 kV L	Reconductor line	\$3,537,000	12% (\$436,000)
Windhub Substation to Dillion 138 kV	Reconductor line	\$624,000	12% (\$77,000)
	Total allocated to project		\$5,509,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.